Basics of Linear Programs

Review Learning Objectives

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1. Consider the following linear program: **1 / 1 point**

$$\begin{aligned} \max \quad & 3x + 4y \\ & x + y \leq 4 \\ & x \geq 0 \\ & y \geq 0 \end{aligned}$$

Check all options that are correct.

- [x] The optimal solution for the above LP is 16.

  Correct
  Correct: The optimal solution for the LP exists at (0,4). Max value of z at this point is 16.

- [ ] If the constraints were changed to : $x + y < 4$, it is still a linear program
- [x] If we remove the constraint

$$x + y \leq 4$$

  it is still a feasible LP problem.

  Correct
  Correct: It is feasible as we can find x,y values that satisfy remaining constraints , but it is unbounded as x,y can take infinitely many values.

- [x] If we changed the problem's objective function to

$$\min -3x - 4y$$

  then the optimal solution for (x,y) remains unchanged.

  Correct
  Correct: Instead of $max(3x + 4y)$ we can minimize it's negation , and get the same answer without loss of generality.

- [ ] (-1,4) is a feasible but not the optimal solution for the above LP probelm.

2. Consider the following scenario. **1 / 1 point**

Green Energy Solutions is a company specializing in solar panel installations. They have a limited workforce and want to maximize their profit while adhering to certain constraints. The company has two types of solar panel installations: Type A and Type B.

Each Type A installation:

- Requires 12 hours of labor
- Costs $500 in materials
- Generates $800 profit per installation.

Each Type B installation:

- Requires 8 hours of labor
- Costs $300 in materials
- Generates $600 profit per installation.

The company has a total of 1000 labor hours available.

The company has $400,000 in cash at hand.

Due to supply constraints, the company can only install a maximum of 300 Type A panels.

Formulate a linear program to maximize the profits while adhering to the constraints above. Let decision variable A denote the number of type A installation and decision variable B denote the number of type B installations.

Choose the correct answers from the choices given below.

( )
$$\begin{aligned} \max \quad & 800A + 600B \\ & 12A + 8B \leq 1000 \\ & 500A + 300B \leq 400000 \\ & A \leq 300 \end{aligned}$$

( )
$$\begin{aligned} \max \quad & A + B \\ & 12A + 8B \leq 1000 \\ & 500A + 300B \leq 400000 \\ & A \leq 300 \\ & A, B \geq 0 \end{aligned}$$

(•)
$$\begin{aligned} \max \quad & 800A + 600B \\ & 12A + 8B \leq 1000 \\ & 500A + 300B \leq 400000 \\ & A \leq 300 \\ & A, B \geq 0 \end{aligned}$$

( )
$$\begin{aligned} \min \quad & 12A + 8B \\ & 12A + 8B \leq 1000 \\ & 500A + 300B \leq 400000 \\ & A \leq 300 \\ & A, B \geq 0 \end{aligned}$$

Correct
Maximizes profit , while maintaining constraints for labour hours , cost of installation , minimum number of panel A installations , and making sure that both type A and B have non-negative values.

3. Select all options that are linear programs. **1 / 1 point**

- [x]
$$\begin{gathered} \max\ 5x + 3y \\ 2x + y \leq 10 \\ 3x + 2y \leq 15 \\ x, y \geq 0 \end{gathered}$$

  Correct
  Correct: All the constraints and the objective function are linear.

- [ ]
$$\begin{gathered} \max\ x + y \\ x^2 + y \leq 10 \\ 2x + 3y \leq 15 \\ x, y \geq 0 \end{gathered}$$

- [ ]
$$\begin{gathered} \max\ 3sin(x) + 4cos(y) \\ 3x + 4y \leq 12 \\ x + y \geq 4 \\ x, y \leq 10 \end{gathered}$$

- [ ]
$$\begin{gathered} \max\ 4x + 3y \\ 2x + y < 8 \\ x + 2y \leq 6 \\ x, y \geq 0 \end{gathered}$$

- [x]
$$\begin{gathered} \min\ 5x + 7y \\ 3x + 2y \leq 12 \\ 4x + 5y \leq 15 \\ x, y \geq 0 \end{gathered}$$

  Correct
  Correct: All the constraints and the objective function are linear.

4. **1 / 1 point**

$$\begin{gathered} \max\ 2x + y \\ x + y \geq 5 \\ x, y \geq 0 \end{gathered}$$

The given linear program is:

(•) Feasible but not bounded
( ) Bounded but not feasible
( ) Neither bounded nor feasible
( ) Bounded and feasible

Correct
The problem is certainly feasible. For instance, $x = 5, y = 0$ is a feasible solution. But it is unbounded: we can have solutions that exceed any desired value of the objective function. For instance, the solution $x = 10^n, y = 10^n$ is a feasible solution but achieves a value of 3×10^n for the objective.

5. Select all the options that are correct. **1 / 1 point**

- [x] An LP problem can have more than one optimal solution

  Correct
  Correct: Different values of decision variables , could give the same optimal value of the objective function.

- [ ] Every LP problem has at least one optimal solution.
- [ ] If the feasible region is not bounded , then the LP problem has no optimal solution
- [ ] Every LP problem has a unique solution.